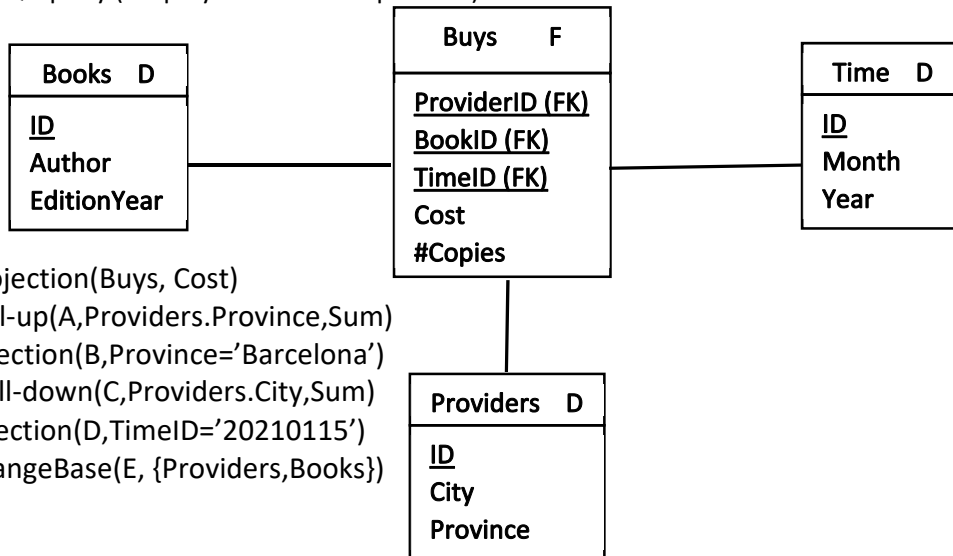


Surname: _____ Name: _____

1)(20%) Given the multidimensional schema and the sequence of **multidimensional operations** below, give the equivalent SQL query (simplify it as much as possible).



- A:=Projection(Buys, Cost)
- B:=Roll-up(A,Providers.Province,Sum)
- C:=Selection(B,Province='Barcelona')
- D:=Drill-down(C,Providers.City,Sum)
- E:=Selection(D,TimeID='20210115')
- R:=ChangeBase(E, {Providers,Books})

2) (25%) Clearly indicate using the line identifiers what would change in the following algorithm if we declare the index over S.A to be a join index.

- for each block of R
 - 1.1. read block of R into a memory page
 - 1.2. for each tuple t in the page read
 - 1.2.1. go through the index of S.A using the value of t.A
 - 1.2.2. if there is any tuple satisfying the join condition
 - 1.2.2.1. go to the corresponding tuples of S
 - 1.2.2.2. generate result
 - 1.2.3. endif
 - 1.3. endForEach
- endForEach

SELECT S.C
FROM R, S
WHERE R.B='x' AND R.A=S.A;

3) (15%) Give an example of query whose syntactic optimization would require the merge of two projections.

Surname: _____ Name: _____

4) (20%) Given two book providers with the database schemas below, write one SQL sentence returning the data in both schemas and loading dimension table BooksDim (below). Specifically: ID, Author, EditionYear, NobelPrize, Genre. Consider that some books can be offered by both providers at the same time. In this case, all the attributes except the Genre should prevail from the Provider1, if available. In case Genre is not available, it should be marked as “Miscellaneous”.

Provider1 schema: <pre>CREATE TABLE P1_Books (Id INTEGER, Author VARCHAR(30), Year CHAR(4), Genre VARCHAR(15), PRIMARY KEY (Id)) CREATE TABLE P1_Nobels (BookId INTEGER, Year CHAR(4), PRIMARY KEY (BookId), FOREIGN KEY (BookId) REFERENCES P1_Books(Id))</pre> Provider2 schema: <pre>CREATE TABEL P2_Books (Id INTEGER, Author VARCHAR(30), Year INTEGER, Nobel BOOLEAN, Genre VARCHAR(15), PRIMARY KEY (Id), CHECK (Year > 0 AND Year <=9999))</pre>	Books Dimension: <pre>CREATE TABLE BookDim (ID INTEGER, Author VARCHAR(30), EditionYear INTEGER, NobelPrize BOOLEAN, Genre VARCHAR(15), PRIMARY KEY (ID))</pre>
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5) (20%) For the case of urban transportation, propose data modeling that enables storing the spatial information about the city bus networks that connect different neighborhoods of the city. In particular, using **MODS ST model**, propose (and briefly justify your answer):

a. **Spatial data types** to model for the following concepts:

i. **bus line** (specifying the line direction),

ii. **city bus network**,

iii. **bus stop**,

iv. **city neighborhood** that the buses trespass.

b. **Spatial operations** that define the **topological** relationships (at least one) between each two of the previous spatial concepts.

i. **bus line - city bus network**:

ii. **bus line - bus stop**:

iii. **bus line - city neighborhood**:

iv. **city bus network - bus stop**:

v. **city bus network - city neighborhood**:

vi. **bus stop - city neighborhood**:
